

Figure 1: Classification of machine learning techniques
(image credits: www.analyticsvidhya.com)

2. **Unsupervised learning:** This is a machine learning technique which is implemented when there is only a set of inputs available with the system, with no corresponding outputs. Now it's left to the system to learn and identify the pattern or rule governing the available inputs by using unsupervised learning and, further, that hypothesis or rule is used to find the output for the given set of inputs. There can be many possible hypotheses, but the optimal one out of all of them is considered for finding the output. Again, the unsupervised learning technique is further classified as K Means and Hierarchical Clustering problems, on the basis of the different techniques used to find the final hypothesis.
3. **Reinforcement learning:** Here the system is given two different sets of input data and it needs to implement the Reinforcement Machine Learning technique in order to learn and identify the general pattern or hypothesis in one of the given set of inputs. There can be more than one hypothesis derived but, finally, the optimal one out of all those available is used by the system to derive the output for the other set of inputs. This is like learning the rules of a game by playing against an opponent.

Implementation of machine learning in real life scenarios

Let's now look at implementing machine learning in real life scenarios. We need to check how we can teach machines to make decisions and do our work just as we would do it, by applying our own sense or logic.

Actually, in the course of teaching machines, every stage of the process helps to build a better version of the machine. There are five basic steps that need to be followed prior to letting a machine perform any unsupervised task.

1. **Collecting data:** This is one of the first and foremost steps in implementing any type of machine learning technique. The data plays quite a significant role in machine learning, whether it is in the form of raw data from MS Excel, Access or even text files. This step lays the foundation of future learning. We must be aware of the fact that the

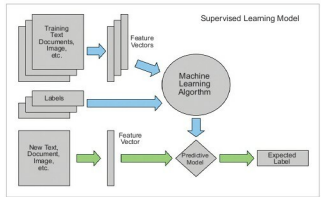


Figure 2: Supervised learning model (image credits: Google images)



Figure 3: Steps to implement machine learning

better the variety, volume and density of relevant data, the better will be the learning prospects for the machine.

2. **Preparing the data:** Once the data is collected, we need to check the quality of what will be fed as training data to the system. We need to spend time in order to determine the quality of data and, accordingly, take steps to fix issues such as treatment of outliers and missing data. Exploratory analysis is one such methodology used to study the differences of the data in detail, thereby strengthening the 'nutritional content' of the data.
3. **Training a model:** This step involves selecting the appropriate algorithm and representing data in the form of a model. The final purified data is split into two parts – *training* and *test* (the proportion of the data depends on the prerequisite requirements). The first part (training data) is used to develop the model, whereas the second part (test data) is used as the reference.
4. **Evaluating the model:** This step involves the evaluation of the machine learning model we chose to implement. The second part of the data (test data) is used to test the accuracy of the learning model. This step actually determines how precise the algorithm selected is, based on the outcome. There is also a better test to check the accuracy of the model, which sees how the model performs on data that has not been used at all while building it.
5. **Improving the performance:** This step may involve choosing a different model altogether or even introducing more variables to improve the efficiency of the learning model. If the model is changed, then it again needs to be evaluated and its performance checked, which is why a lot of time needs to be spent in collecting and preparing data.